

X-ray Crystallography

Our high-school understanding suggests that within crystalline materials, atoms are arranged periodically with a certain relationship with their neighbours. Discussion on periodic arrangement of points in space comes under the domain of crystallography which provides theoretical models of atomic arrangement in crystalline solids. To physically prove the internal structure of crystalline materials vis-à-vis arrangement of atoms and their mutual correlation, x-ray diffraction is perhaps the most effective technique. Therefore, the objective of the present discussion is to “understand and describe the internal structure vis-à-vis atomic arrangement of crystalline materials at unit cell level” using x-ray diffraction. To begin with, let us brush up our fundamentals on crystallography.

Fundamentals of crystallography

Atomic arrangement within crystalline materials essentially follows certain structural models. X-ray diffraction and allied techniques are primarily used for analysing such models possessed by crystalline materials. These structural models which we often refer as crystal structures are crucial in deciding the functionalities of associated materials and therefore requires thorough understanding. Development of such understanding comes under the field of crystallography. While a detailed discussion on crystallography is beyond the scope of the current book, we will systematically proceed in understanding elementary crystallography which will be essential for appreciating x-ray diffraction and other techniques discussed in this book, for structural characterization of materials. In the process, we will keep ourselves restricted within the discussion on lattice, motifs, crystals, seven crystal systems and fourteen Bravais lattices. We will begin with the high school level definition of different crystal systems on the basis of unit cell dimensions. We will demonstrate the issues with such definitions and introduce the concept of crystallographic symmetry. We will discuss different symmetry elements and classify lattice on the basis of symmetry. Further we will define “motif” or “basis”. How a crystal is built up upon attaching a basis to a lattice and consequent change in symmetry to the crystal, if any, will be discussed further. We will briefly introduce the important concepts of stereographic projection and reciprocal lattice which are vital to the understanding of several important characterization techniques.

Box 4.1

$$\text{Crystal} = \text{Lattice} + \text{Basis (motif)}$$

Lattice and its classification

Concept of lattice has nothing to do with a crystal. It is a mathematical construction of arranging points in space. To formally define, periodic arrangement of points with translation symmetry in space is termed lattice. Figure 4.1 shows an array of points arranged in one dimension (1D) where each point is separated from its neighbour by equal distance. If we extend these points in both positive and negative directions till infinity, the array of periodically arranged points form a one- dimensional lattice. The position vector (lattice vector) of each point here can be defined by:

$$\vec{R} = n\vec{a} \quad (1.1)$$

where n are integers and assume values both positive and negative. $n = 0, \pm 1, \pm 2$, $\vec{a}$ is a vector, termed primitive translation vector. Magnitude of $|\vec{a}|$ is called lattice parameter. This implies that taking any lattice point as the origin, one can generate other lattice points and thus the entire lattice by choosing all imaginable integer values of n and therefore having different lattice vectors. Thus, the array of points with such translational symmetry, shown in Fig. 4.1 fulfils the definition of a one dimensional (1D) lattice. It should further be noted that within 1D space we cannot make further classifications of lattice unlike two-dimensional (2D) and three dimensional space (3D) discussed below.



Fig. 4.1: Periodic, infinite arrangement of points in one-dimension constitutes one-dimensional lattice

Two-dimensional lattice: Within two dimension, however, one can imagine different kinds of periodic arrangement of points vis-à-vis lattices. Figure 4.2 schematically demonstrates all the possible types of 2D lattices; (a) oblique, (b) rectangular, (c) centered rectangular, (d) hexagonal and (e) square. Similar to 1D lattice, 2D lattice possess translational symmetry. However, one requires two primitive translation vectors to define a lattice vector and the 2D lattice:

$$\vec{R} = m\vec{a}_1 + n\vec{a}_2 \quad (1.2)$$

where $\vec{a}_1$ and $\vec{a}_2$ are the primitive translation vectors and m and n are integers which can

assume any combination of integer values, both positive and negative. The magnitude of $|\vec{a}_1|$ and $|\vec{a}_2|$ represent lattice parameters. In 2D, the angle (ϕ) between the lattice vectors, $\vec{a}_1$ and $\vec{a}_2$ constitute the third lattice parameter. Conventionally, the correlation and magnitude of the lattice parameters ($|\vec{a}_1|$, $|\vec{a}_2|$ and ϕ) are used to classify the lattices in 2D.

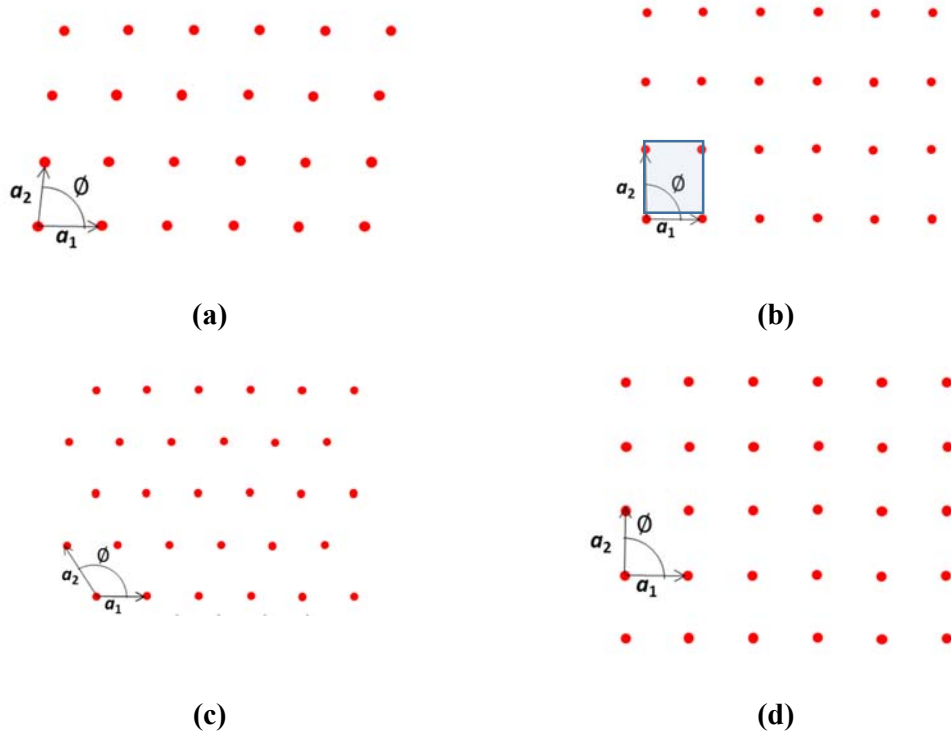


Fig.4.2: Two dimensional lattices, viz., (a) oblique, (b) rectangular, (c) centered rectangular, (d) hexagonal and (e) cubic arranged in increasing symmetries.

Oblique lattice is defined as the one where the correlations between the lattice parameters are: $|\vec{a}_1| \neq |\vec{a}_2|$ and $\phi \neq 90^\circ$. The periodic arrangement of points within an oblique lattice is demonstrated in Fig. 1.2(a). It is noticed that this lattice has the highest degree of freedom and least symmetry albeit possessing the translational symmetry which is the defining character of a lattice. If one imposes some restriction such that: $|\vec{a}_1| \neq |\vec{a}_2|$ but $\phi = 90^\circ$, the resultant lattice is termed rectangular lattice, as shown in Fig. 4.2(b). Using the two lattice vectors, one can construct a rectangle, as shown in Fig. 1.2(b) and thus the name rectangular lattice arises. This rectangle also possesses translational symmetry i.e., if repeated in 2D space, it can generate the entire lattice. This rectangle is called “unit cell” which is the basic building block of a lattice. We will come back to the discussion of “unit cell” later. For hexagonal

lattice, $|\mathbf{a}_1| = |\mathbf{a}_2|$ but $\phi = 120^\circ$ (Fig. 4.2(d)). A stack of the hexagonal unit cells generate a hexagon. When $|\mathbf{a}_1| = |\mathbf{a}_2|$ but $\phi = 90^\circ$, one gets a cubic lattice. The above classifications are summarized in Table 1.1.

Table 4.1: Classification of 2D lattices.

SI No.	Two-dimensional lattice	Unit cell shape
1	Oblique	$a \neq b ; \phi \neq 90^\circ$
2	Rectangular	$a \neq b ; \phi = 90^\circ$
3	Hexagonal	$a = b ; \phi = 120^\circ$
4	Square	$a = b ; \phi = 90^\circ$

Three-dimensional lattice: In three dimensions (3D), in commensurate with one- and two-dimensions, one requires three translation vectors to define the lattice:

$$\vec{R} = m\vec{a}_1 + n\vec{a}_2 + p\vec{a}_3 \quad (1.3)$$

where $\vec{a}_1$, $\vec{a}_2$ and $\vec{a}_3$ are the primitive translation vectors and m , n and p are integers which can assume any combination of integer values, both positive and negative. The magnitude of $|\vec{a}_1|$, $|\vec{a}_2|$ and $|\vec{a}_3|$ represent lattice parameters. In 3D, the angles α, β and γ between the translation vectors, $\vec{a}_2 \wedge \vec{a}_3$, $\vec{a}_1 \wedge \vec{a}_3$, and $\vec{a}_1 \wedge \vec{a}_2$, respectively constitute the three more lattice parameters. Conventionally, the correlation and magnitude of the lattice parameters $|\vec{a}_1|$, $|\vec{a}_2|$, $|\vec{a}_3|$, α, β and γ are used for classification of the lattices in 3D. There are seven different types of three- dimensional lattices also termed seven crystal systems, viz., triclinic, monoclinic, rhombohedral, tetragonal, hexagonal, orthorhombic and cubic, respectively.

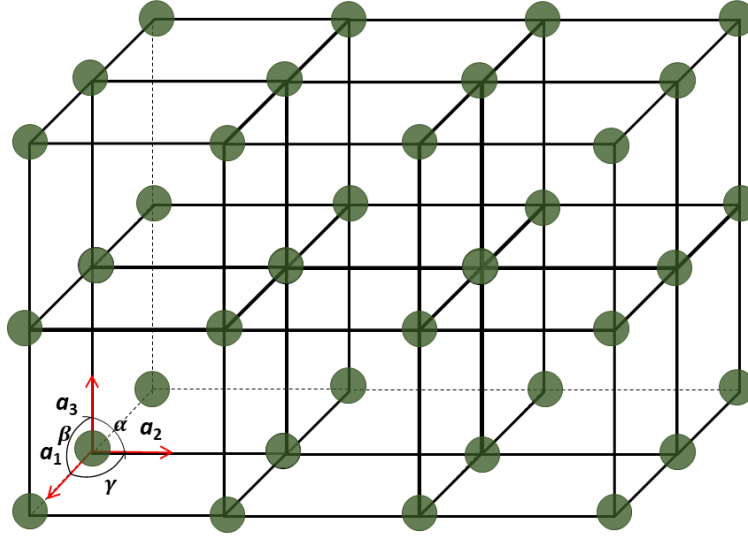


Fig. 4.4: a schematic of a three dimensional lattice.

Table 4.2 lists the usual definitions of all seven three dimensional lattices/crystal systems and Fig. 4.5 shows the schematics of the conventional unit cells corresponding to each system. It is observed that the triclinic system has the highest degrees of freedom i.e, least symmetry and the cubic has the highest symmetry. The readers may note the term “symmetry” here which in the subsequent sections will be elaborated further.

Table 4.2: Classification of 3D lattices.

SI No.	Crystal systems	Unit cell shape
1	Triclinic	$a \neq b \neq c ; \alpha \neq \beta \neq \gamma$
2	Monoclinic	$a \neq b \neq c ; \alpha = \beta = 90^\circ \neq \gamma$
3	Rhombohedral	$a = b = c ; \alpha = \beta = \gamma \neq 90^\circ$
4	Hexagonal	$a = b \neq c ; \alpha = \beta = 90^\circ ; \gamma = 120^\circ$
5	Orthorhombic	$a \neq b \neq c ; \alpha = \beta = \gamma = 90^\circ$
6	Tetragonal	$a = b \neq c ; \alpha = \beta = \gamma = 90^\circ$
7	Cubic	$a = b = c ; \alpha = \beta = \gamma = 90^\circ$

Before we move further, it would be pertinent to clarify few terms which we have included in Table 4.1 and 4.2 subsequent discussions without really explaining them for instance unit cell and Bravais lattice.

4.1.1.1 Unit Cell

Defined by lattice vectors, as shown in Figure 4.5(a), there are infinite possibility of constructing parallelograms based on infinite combination of possible translation vectors. In the two dimensional space, each of these parallelograms is defined as unit cell. As seen in the figure, for a given lattice, the choice of unit cell is numerous similar to the choices of translation vectors. Unit cells in a lattice can be broadly classified into two classes: (i) primitive and (ii) non-primitive or conventional unit cells. Primitive unit cell of a Bravais lattice is the smallest building block with “one” effective lattice point per cell that can envelope the entire lattice upon translation without overlapping. As demonstrated in Figure 4.5(b), all the unit cells qualify the definition of primitive cell. The only criterion that should be ensured is that the unit cell should have not more than one effective lattice point. At this juncture it should be mentioned that though the choice of primitive unit cell is not unique the area/volume enveloped in 2D/3D, respectively by all the primitive cells for a given lattice is identical. This could be easily verified by computing the area of the two unit cells shaded in Figure 4.5. Let us assume that the magnitude of the translation vectors defining the lattice are: $|\vec{a}_1| = a$ and $|\vec{a}_2| = b$. The unit vectors are $\frac{\vec{a}_1}{|\vec{a}_1|} = \hat{x}$ and $\frac{\vec{a}_2}{|\vec{a}_2|} = \hat{y}$, respectively. Therefore, one can rewrite, $\vec{a}_1 = a\hat{x}$ and $\vec{a}_2 = b\hat{y}$. With these, one can further define $\vec{a}_3 = \vec{a}_1 + \vec{a}_2 = a\hat{x} + b\hat{y}$. Area of the primitive unit cells enclosed by $\vec{a}_1, \vec{a}_2$ and $\vec{a}_1, \vec{a}_3$ are as follows:

$$\Omega_A = |\vec{a}_1 \times \vec{a}_2| = ab \sin \frac{\pi}{2} = ab \text{ and}$$

$$\Omega_B = |\vec{a}_1 \times \vec{a}_3| = |\vec{a}_1 \times \vec{a}_3| = a\hat{x} \times (a\hat{x} + b\hat{y}) = ab$$

Similar approaches could be attempted to demonstrate that all other primitive unit cells have identical area. This could be further validated in 3D as well.

Though choice of primitive unit cell is not unique, associated with every Bravais lattice there is one unique primitive unit cell, named Wigner-Seitz (WS) cell. The speciality of WS cell is that it consists of the region of space around each lattice point of the Bravais lattice where all points within this region is closer to the above mentioned lattice point than any other lattice point belonging to the lattice under consideration. Let us consider the same lattice as shown in Fig. 4.5. Let us focus on a particular lattice point and join it with the

nearby lattice points. Take perpendicular bisectors on each of these straight lines, as shown in Fig. 4.6. Join them. The smallest area enclosed is the Wigner-Seitz cell corresponding to the given lattice. Similar approach in three-dimensions produces three dimensional Wigner-Seitz cell. At this juncture, it should be noted that one can construct WS cell for any lattice including reciprocal lattices. Wigner-Seitz cell of a reciprocal lattice results in the first Brillouin zone corresponding to the lattice. A detailed discussion on this is beyond the scope of the present text while we will return to the discussion of reciprocal lattice in section 4.xx.

Non-primitive (sometimes conventional) cells, on the other hand possess more than one lattice points. Typical non-primitive cells in three dimensions with cubic structure includes body centered cubic and face centered cubic unit cells. One can derive a primitive unit cell out of the conventional unit cell. The above discussion is summarized in Box 4.2.

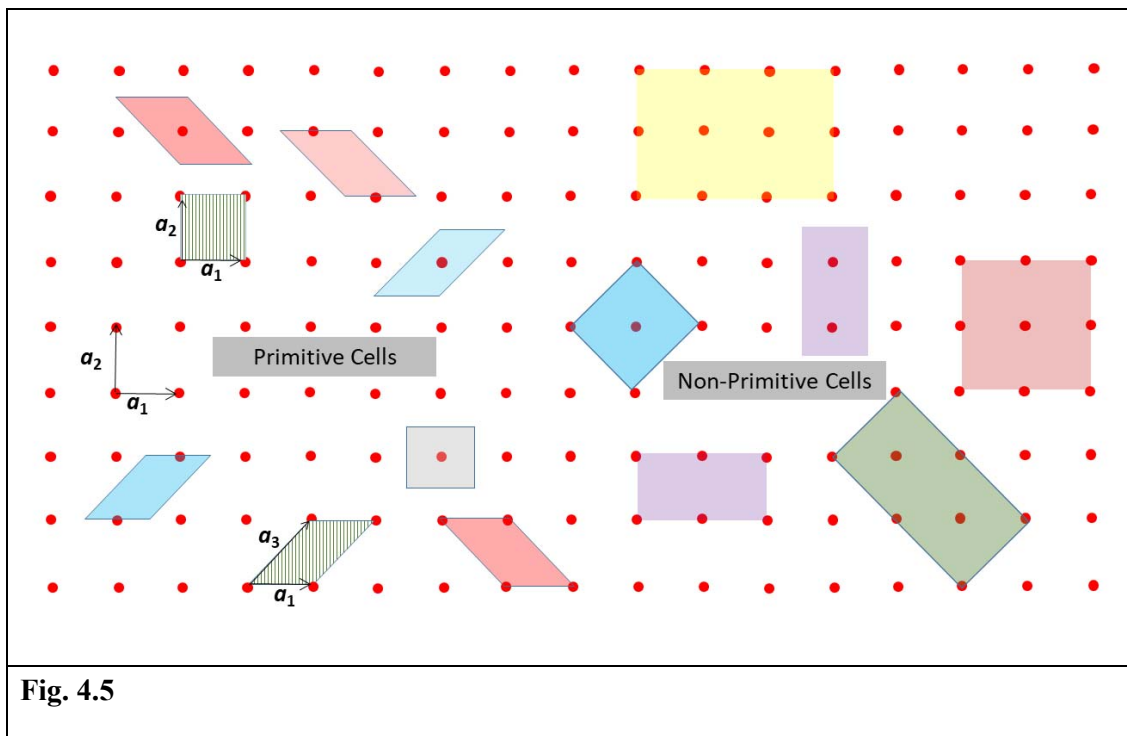
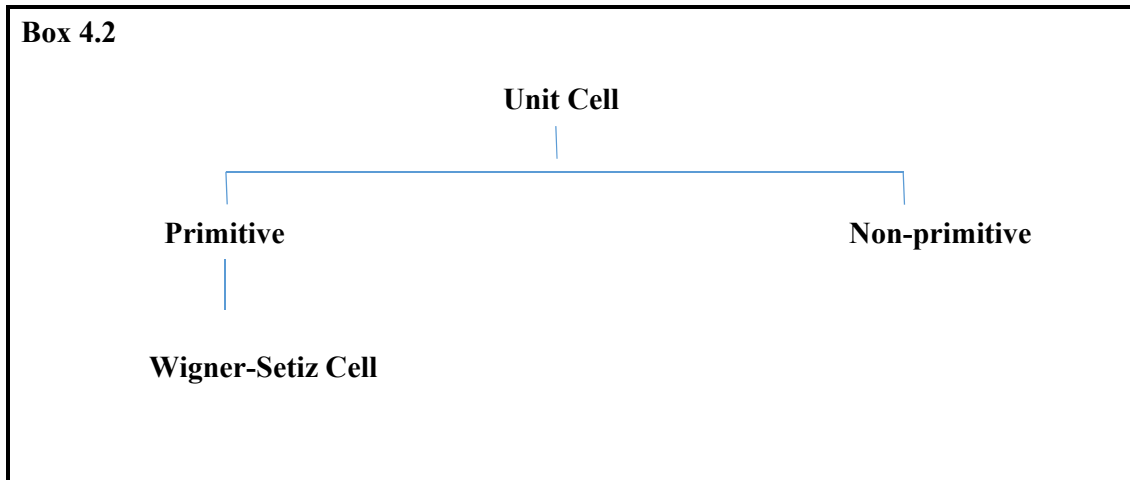
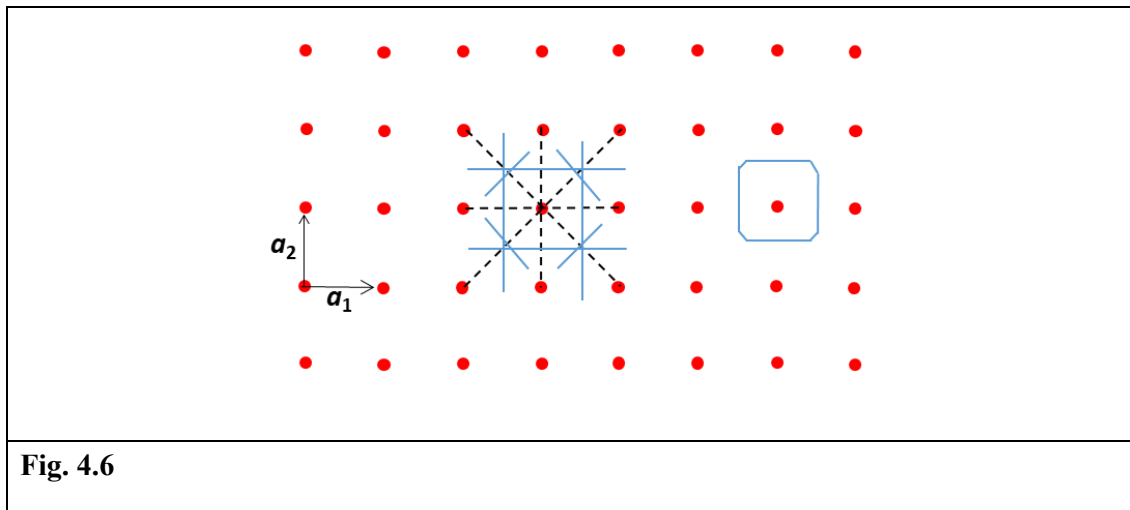


Fig. 4.5



4.1.1.2 Bravais lattice

As the name suggests, Bravais lattice is a type of lattice. Therefore, Bravais lattice is a mathematical construction of infinite, periodic arrangement of points in n -dimensional space where each point has identical surroundings. In the current discussion on solid state crystallography, we will restrict ourselves up to three dimensions. All the points in a (Bravais) lattice can be defined by appropriately chosen primitive translation vectors. The translation vectors in 2 dimensions' form a parallelogram as shown in Figure 4.4 (a) and construct a parallelepiped in three dimensions. It can be observed that if one repeats the parallelogram/parallelepiped in space it can tile the entire space without any overlapping. Figure 4.4 (a) also shows that the choice of the translation vector is not unique. In fact, there could be an infinite combination of the translation vectors, as shown in Figure 4.4 (a) and each combination can

result in parallelograms which can tile the space. At this juncture it will be worth mentioning the difference of Bravais lattice from non-Bravais lattice. Figure 4.4(b) shows a long range arrangement of points. The arrangement is periodic and thus qualifies the definition of a lattice. However, it noticed that in this lattice, not all points have identical surroundings. For example, point A and point B in Figure 4.4(b) have different surroundings. Thus, Figure 4.4(b) does not define a Bravais lattice. In other words, in case of non-Bravais lattice, one or more number of lattice points are non-equivalent. On the contrary, atomic arrangement in Figure 4.4 (a) corresponds to the definition of Bravais lattice. Therefore, all Bravais lattices are lattices but the converse may not be true. In this book, unless otherwise mentioned, we would treat “lattice” as “Bravais lattice”.

Bravais lattice in one- dimension does not have any sub-classifications for obvious reason. All 1D Bravais lattices are primitive (P) lattice. In two dimensions, however, one can have both primitive (P) as well as non-primitive/centered (C) lattices. Within the unit cell, if there lies a lattice point, the lattice is termed a centered lattice (C). Table 4.3 lists all the Bravais lattices in 2D lattices or crystal systems. It is noticed that not all the two- dimensional crystal systems possess both primitive and centered lattices. Within the rectangular unit cell, if one imagines a lattice point at the center, the resultant lattice is called centered rectangular (Fig. 4.2 (c)).

Table 4.3: 2D Bravais lattices.

Sl No.	Two-dimensional lattice	Unit cell shape	Bravais Lattice	
			Primitive (P)	Centered (C)
1	Oblique	$a \neq b ; \phi \neq 90^\circ$	P	
2	Rectangular	$a \neq b ; \phi = 90^\circ$	P	C
3	Hexagonal	$a = b ; \phi = 120^\circ$	P	
4	Square	$a = b ; \phi = 90^\circ$	P	

Table 4.3 introduces few important questions towards the classifications of 2D lattices. For instance, it is observed that rectangular lattice has a sub-variant called centered-rectangular but others do not have. One can raise a very pertinent question, why? Why, for example, square

lattice does not have a centered variant alike rectangular one? A tentative albeit not exact argument we tend to present is as follows:

Figure 4.3 (a) present a square lattice. An attempt to center the lattice with a centered unit cell is shown which leaves a centered cubic unit cell. However, it is observed that one can create a simple square unit cell out of the centered square unit cell which would be smaller in dimension as shown in the same figure. Since primitive-square is already in the list, there is no point to add centered square as a separate entity. A similar attempt on centered rectangular cell, however, does not produce a simple rectangular cell and therefore centered rectangular makes into the list. Centered hexagonal is not included in the list of Bravais lattices in 2D. In Fig. 4.3 (b), an attempt to produce a centered hexagonal cell is demonstrated. One cannot produce a simple hexagonal with smaller volume from a centered hexagonal cell unlike its cubic counterpart. As per the above argument, it should have been included in the list of Bravais lattices. However, here the traditional argument is, well you cannot produce a primitive hexagonal cell from a centered hexagonal but you can produce a primitive rectangular cell with smaller volume and therefore centered hexagonal is redundant. However, the same argument should have been applied to centered rectangular cell where one can generate a primitive oblique cell with smaller volume! Conventional argument is silent here. Further, the classification further does not clarify why we cannot have a Bravais lattice with $|a_1| = |a_2|$ and $\angle \neq 90^\circ$.

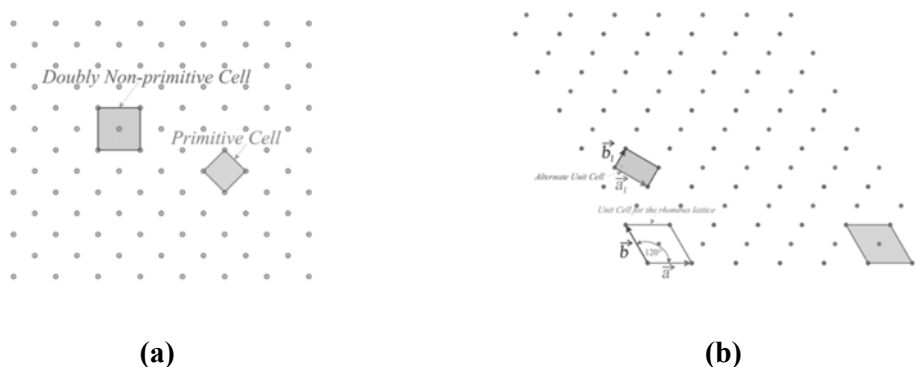


Fig. 4.3:

In three-dimensions, as we see in Table 4.4, there are four different possible variants of Bravais lattices viz., primitive (P), body centered (I), face centered (F), base centered (A/B/C depending on the centering) Fig. 4.xx shows the unit cells of the above four variants. Not all crystal systems possess all the above four different variants of bravais lattices. For instance, in case of cubic system, one finds three different types of Bravais lattices viz., primitive (P), body

centered (I) and face centered (F). One can ask why is there no base centered (C) cubic cell? A possible explanation put forward is as demonstrated in Fig. 4.xx:

In Fig. 4.xx, there are two edge- centered cubic unit cells are considered. Each of these cells correspond to conventional cubic unit cell. However, it is noticed that if one joins the points at the base centers and the corners (shaded in green), one can envelope a cell outlined in blue. This new cell is also a unit cell because if repeated in space it can tile the entire lattice. Using simple geometry, one can find that the shape of this new unit cell corresponds to the dimensions of a tetragonal crystal system. The new unit cell which is a primitive tetragonal cell possesses lower volume than that of the original base centered cell and therefore makes into the list of Bravais lattice. In summary, base- centered cubic and primitive tetragonal cells are equivalent. The later having smaller volume makes into the list of Bravais lattice.

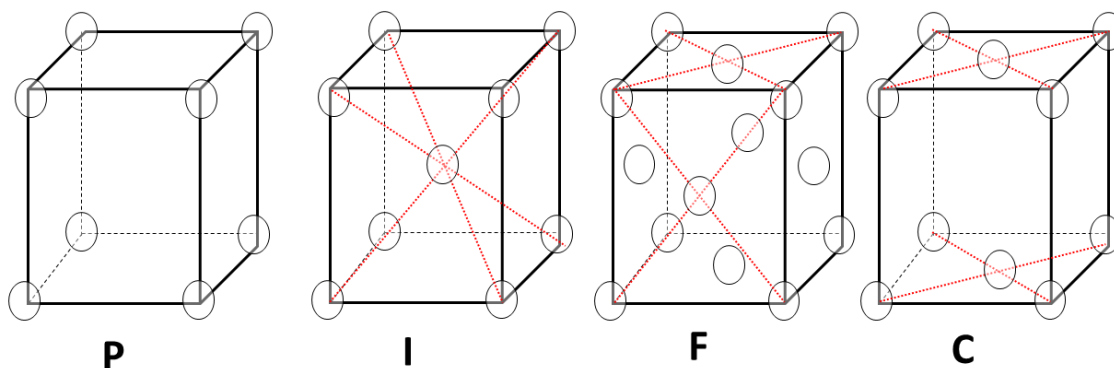


Fig. 4.xx

Table 4.2: Classification of 3D lattices.

Sl No.	Crystal systems	Unit cell shape	Bravais Lattice			
			P	I	F	A/B/C
1	Triclinic	$a \neq b \neq c ; \alpha \neq \beta \neq \gamma$	P			
2	Monoclinic	$a \neq b \neq c ; \alpha = \beta = 90^\circ \neq \gamma$	P			C
3	Rhombohedral	$a = b = c ; \alpha = \beta = \gamma \neq 90^\circ$	P			
4	Hexagonal	$a = b \neq c ; \alpha = \beta = 90^\circ ; \gamma = 120^\circ$	P			

5	Orthorhombic	$a \neq b \neq c ; \alpha = \beta = \gamma = 90^\circ$	P	I	F	C
6	Tetragonal	$a = b \neq c ; \alpha = \beta = \gamma = 90^\circ$	P	I		
7	Cubic	$a = b = c ; \alpha = \beta = \gamma = 90^\circ$	P	I	F	

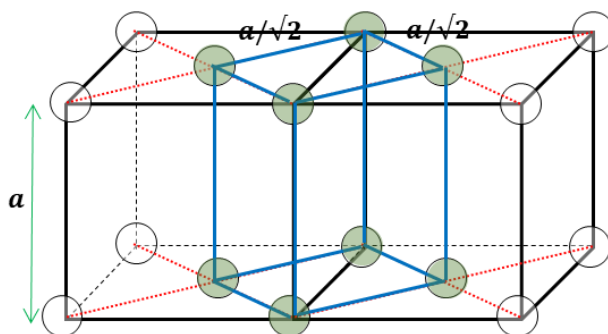


Fig. 4.xx

Another similar question could be why do not we have a body centered monoclinic lattice? To understand this let us draw two body centered monoclinic cell side-by-side. It is observed that these two cells let one construct a base centered monoclinic cell, as shown in Fig. 4.xx(b). Since both the cells are equivalent, it would be sufficient to consider one of them in the list of Bravais lattice.

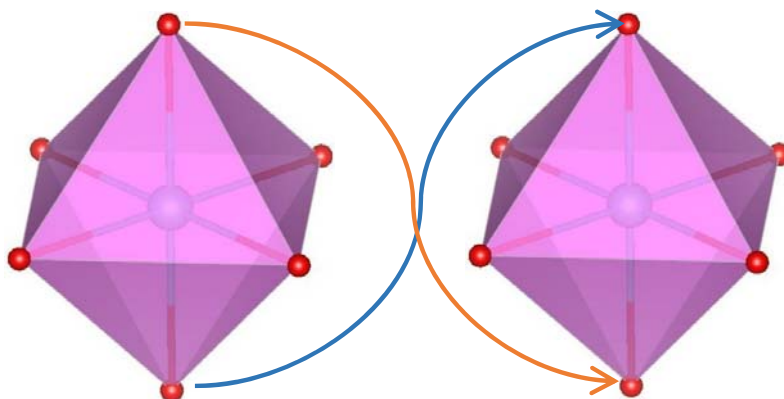
Let us apply identical argument on face centered cubic lattice. Let us consider two adjacent cells as shown in Fig. 4.xx(c). It is observed that using similar approach, one can construct a body centered tetragonal cell which is smaller in volume than the original cubic cell. However, we see that both body centered tetragonal and face centered cubic make into the list of Bravais lattice. What could be the reason? In order to explain these apparent inconsistencies, the traditional definitions of the crystal systems that we defined so far is to be questioned. Let us revisit Fig. 4.5. We have seen that for a lattice, the choice of unit cell is infinite. One can construct different unit cells of different shape and sizes, corresponding a given lattice. Thus, if the choice of unit cell for a given lattice is not unique, how could one classify Bravais lattices based on the dimensions of the unit cell? In fact, in three dimension, $a = b = c ; \alpha = \beta = \gamma = 90^\circ$ should not be the definition of a cubic lattice because one can construct even a triclinic cell on the same lattice. If one defines the lattice based on the dimensions of this triclinic cell and mark the lattice triclinic, that will not be appropriate. In

general, unit cell dimensions does not necessarily warrant the specific crystal class. Then, what defines crystal classes and Bravais lattices? In this regard, “symmetry” should be the basis of classification of Bravais lattice. Let us understand what do we mean by the term, “symmetry”.

4.1.1.3 Symmetry

If an object or a pattern or a lattice is brought into self-coincidence after some operation, the object or the pattern or the lattice is said to possess symmetry with respect to that operation. It means that the object/pattern/lattice will appear identical before and after the symmetry operation. In other words, the object/pattern/lattice remains invariant upon a symmetry operation. Let us take an example. Refer to Fig. 4.5 which represents a two dimensional square lattice. Now, consider any lattice point on this lattice. Carefully note the surroundings of the above point. Now, if we apply translation operation, defined by eq. (1.2), we move to a new lattice point. If one evaluates the surroundings of this new lattice point, it is identical to the earlier situation. There is absolutely no difference so much so that one cannot distinguish the new point, separately. Considering it is infinite, the lattice remains invariant after the above translation operation. Thus, the above translation operation is a symmetry operation and the lattice possesses translational symmetry. It is the property of Bravais lattices that they possess translational symmetry. Bravais lattice, in addition to the translational symmetry, can further possess other types of symmetries. These symmetry operations can be operated about a point or an axis or a plane. The point, the axis or the plane about which a symmetry operation is performed is termed the symmetry element. Crystal lattice can have one or more of the following symmetry elements: (i) center of symmetry, (ii) planes of symmetry, (iii) axes of symmetry, (iv) inversion axes, (v) screw axes and (vi) glide planes.

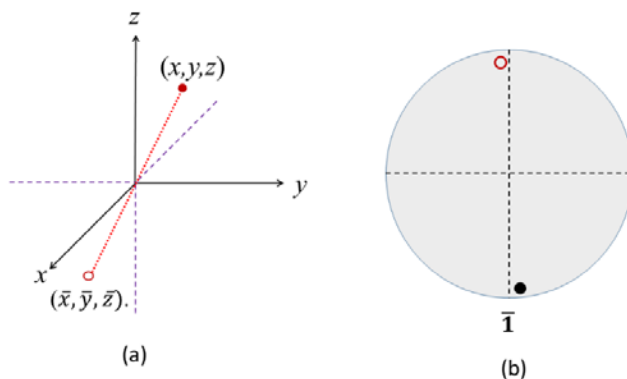
(i) *Center of symmetry* ($\bar{1}$): If an object or a pattern or a lattice is brought into self-coincidence upon inversion, the object or the pattern or the lattice is said to possess a center of symmetry or inversion symmetry. Center of symmetry is represented by $\bar{1}$. To appreciate the concept of inversion symmetry let us consider a regular octahedron with six red spheres at the vertices, as shown in Fig. 4.10(a). If the octahedron is inverted about its center (shown in blue), it will remain identical and one cannot distinguish it from its original configuration and the octahedron would be said to have a center of symmetry. However, if someone replaces one of the spheres at the vertices and substitute with one having a different color (Fig. 4.10(b)), upon inversion, the octahedron will have a different configuration and one can distinguish the difference. Here, the octahedron will be termed as one lacking an inversion center or a center of symmetry.



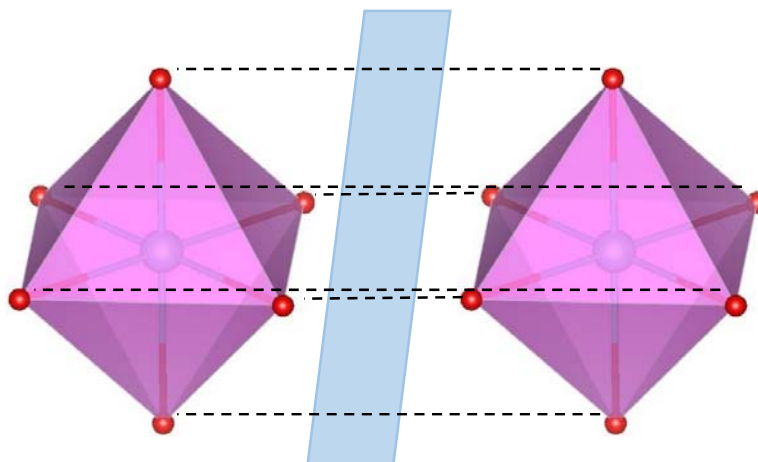
To further demonstrate, a point with coordinate (x,y,z) on a three-dimensional coordinate system will produce an equivalent point due to the operation of the inversion about a point of reference. This is demonstrated in Fig. 4.xx(a). Clearly, the coordinate of the equivalent point is $(\bar{x}, \bar{y}, \bar{z})$. One can also arrive at the coordinate of the above equivalent point by performing a

coordinate transformation where the transformation matrix is: $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$. To

demonstrate symmetry elements and symmetry operations, there is a beautiful representation scheme (stereograms) which shows the generation of equivalent point(s) due to the operation of a symmetry element. Fig. 4.xx(b) shows the stereogram of inversion axis ($\bar{1}$).



(ii) *Planes of symmetry*: A plane of symmetry is a reflection plane which leaves an object or a pattern or a lattice invariant following a mirror reflection operation. It is represented by the letter, m .

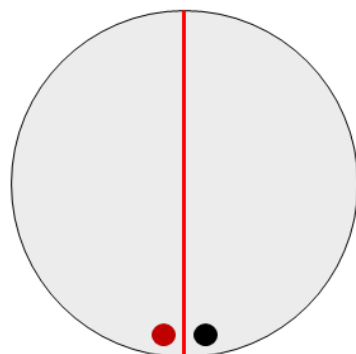


To demonstrate the operation of plane of symmetry or mirror plane (m) let us consider Fig. 4.xx(a). Here, the mirror plane is perpendicular to the x -axis. As a result, a point with coordinate (x,y) , due to the operation of the above mirror plane will produce an equivalent point with coordinate $(\bar{x}, y)$. Here, one needs to note carefully, the location of the mirror plane, had the mirror plane been perpendicular to the y -axis the equivalent point would have the coordinate, $(x, \bar{y})$. Thus, similar to the case of inversion center, one can construct the transformation matrix, which would produce the equivalent point due to the operation of a mirror plane based on its orientation. Table 4.xx, list the equivalent points and transformation matrices for mirror planes oriented along different directions. The generation of the equivalent points is demonstrated in the stereogram, as shown in Fig. 4.xx. It is to be noted that when mirror plane is contained by the plane of the paper, it is represented by $1/m$ and the equivalent point produced lies beneath the plane of the paper and marked by an open circle in the stereogram. When the mirror plane lies perpendicular to the plane of the paper, it is represented by m ; however, the equivalent point is marked in filled circle in the stereogram.

Table 4.xx:

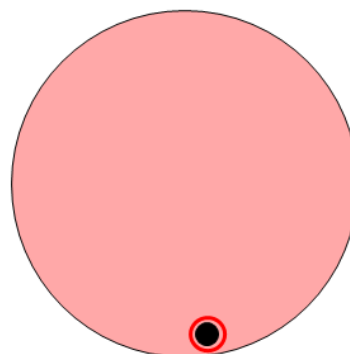
Mirror plane	Perpendicular to	Equivalent points	Matrix
M	a -axis	$(x, y, z), (\bar{x}, y, z)$	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
	b -axis	$(x, y, z), (x, \bar{y}, z)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

c-axis	$(x, y, z), (x, y, \bar{z})$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$
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m

(a)



$\frac{1}{m}$

(b)

Fig. xx:

(iii) *Axes of symmetry*: If an object or a pattern or a lattice remains invariant upon rotation through smallest, non-zero angle, θ , then the object or a pattern or a lattice is said to possess an n -fold axis of rotation where $n = \frac{360^\circ}{\theta}$. The value of n determines the degree of axis. Table 4.3 lists the rotation axes.

Sl No.	Rotation axis (n -fold)	Angle of rotation (θ)	Graphical symbol
1	1	360°	-
2	2	180°	◐
3	3	120°	▲
4	4	90°	■
5	5	72°	⬠

Table	6	6	60°		4.3:
	7	8	45°		

However, in case of Bravais lattice with translational symmetry, 5-fold axis and rotation axes higher than six are not possible. This is because, one cannot tile space by translating a pentagon. There are open space, as shown in Fig. 4.xx. Further, polygons with sides more than six, when translated in space, overlaps on each other, as shown in Fig. 4.xx.

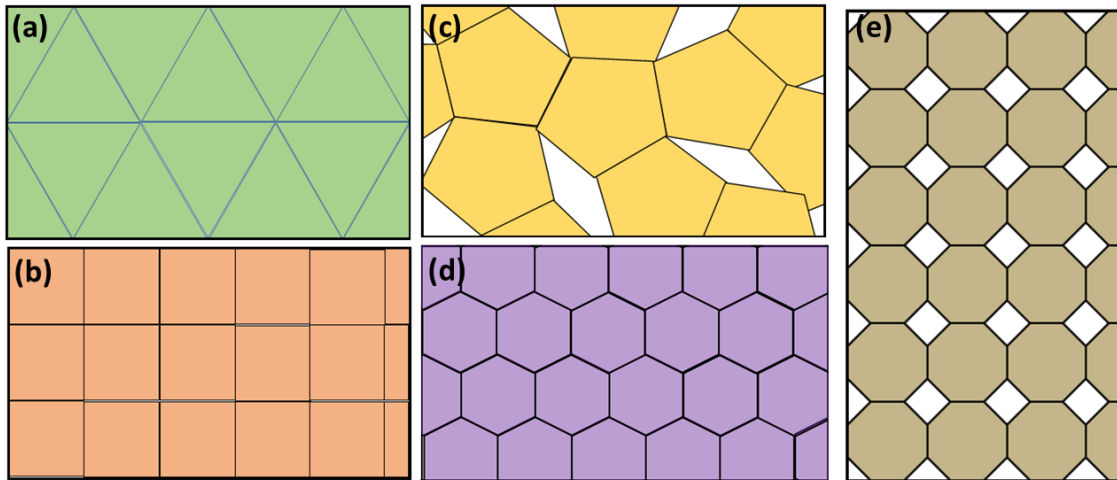


Fig.4.xx

The above restrictions on the rotational symmetry elements in case of Bravais lattices can be proved analytically as follows:

Let us consider a Bravais lattice, with points arranged periodically in space as shown in Fig. 4.11. Distance between the lattice points (lattice parameter) is “ a ”. We will examine the restrictions on the rotational symmetry elements present in the lattice (operating on each lattice point in the lattice). Let us concentrate on point N. Imagine, there is an n -fold axis operating such that $n = \frac{360^\circ}{\theta}$; where θ is the minimum rotation angle upon which the lattice reaches to self-coincidence.

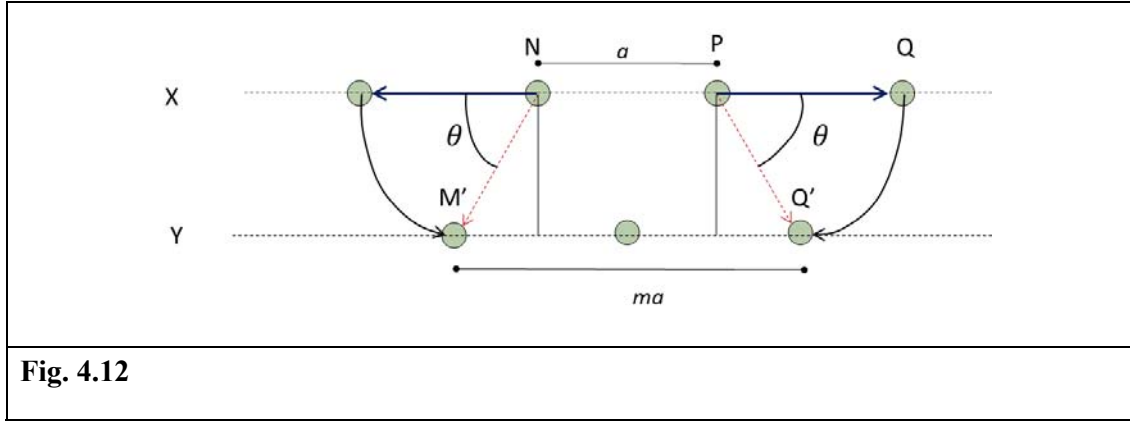


Fig. 4.12

It suggests that the point M in Fig. 4.xx upon rotation through θ in the anti-clockwise manner, about the point N will generate an equivalent point, M' . Let us consider another point P about which we will perform the rotation through θ , however, this time, in the clockwise manner such that a point Q generates an equivalent point Q' . An allowed rotation axis will generate, equivalent points that are part of the lattice. Therefore, if points M' and Q' are two lattice points the rotation axis n is allowed for the given lattice.

It is easy to reason that the line joining the lattice points M and Q is parallel to the one joining M' and Q' . If the points M' and Q' are two lattice points, the distance between them would be integral multiple of the lattice parameter, ma . Therefore, the distance between the two points,

$$M'Q' = M'N' + N'P' + P'Q'$$

$$\text{Or, } ma = a\cos\theta + a + a\cos\theta$$

$$\text{Or, } (m - 1)a = 2a\cos\theta$$

$$\text{Or, } \cos\theta = \frac{(m-1)}{2}$$

While m can assume integer values (both positive and negative), it is restricted by the magnitude of $\cos\theta$ which can only take values, $-1 \leq \cos\theta \leq 1$. Thus, one can determine the allowed values of m , θ and therefore n . Table 4.xx lists these values.

Table 4.xx

Sl. No.	$(m - 1)$	$\cos\theta = \frac{(m - 1)}{2}$	$\theta = \cos^{-1} \frac{(m - 1)}{2}$	$n = \frac{2\pi}{\theta}$
---------	-----------	----------------------------------	--	---------------------------

1	-2	-1	π	2
2	-1	$-\frac{1}{2}$	$\frac{2\pi}{3}$	3
3	0	0	$\frac{\pi}{2}$	4
4	1	$\frac{1}{2}$	$\frac{\pi}{3}$	6
5	2	1	2π	1

Thus, Table 4.xx rules out the possibility of 5-fold as well as higher than 6-fold axes in Bravais lattice.

Let us examine the generation of equivalent points as a result of the operation of the different rotation axes. Let us consider Fig.4.xx(a) and (b). In Fig. 4.xx(a), a 2-fold axis is operating along z-axis. A point with coordinate (x, y) will result an equivalent point, $(\bar{x}, \bar{y})$. It is understood that the coordinates of the equivalent points will depend on the orientation of the 2-fold axis. Table 4.xx lists the equivalent points and transformation matrices for 2-fold axis.

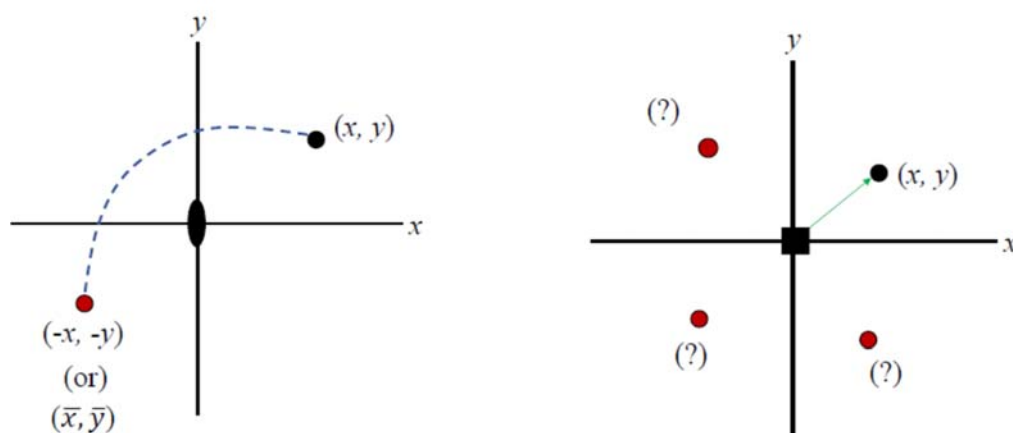
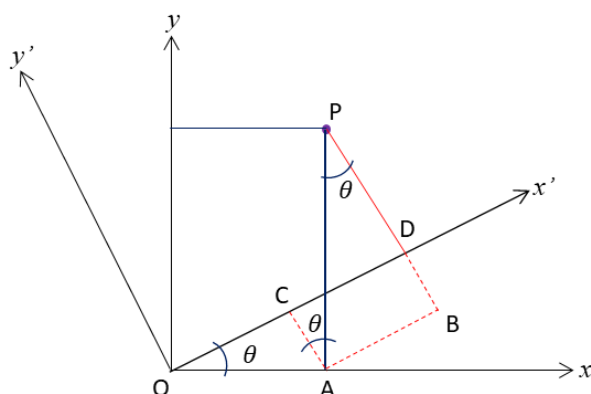


Fig. 4.xx:

Let us do similar exercise for 4-fold axis. There will be four equivalent point generated due to the action of the 4-fold axis operating perpendicular to the plane of paper i.e., along z-axis. To find out the coordinates of all these points, we will recapitulate coordinate transformation summarized in Box. 4.x. From Box 4.xx it is easy to calculate the equivalent points generated due to the above 4-fold rotation.

n-fold axis	Symbol	Orientation	Equivalent points	Transformation matrix
2		along a-axis	$(x, y, z), (x, \bar{y}, \bar{z})$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$
		along b-axis	$(x, y, z), (\bar{x}, y, \bar{z})$	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$
		along c-axis	$(x, y, z), (\bar{x}, \bar{y}, z)$	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Box 4.xx



Let xy Cartesian coordinate system be rotated by an angle θ . The resultant coordinate system becomes $x'y'$. A point P with coordinates (x, y) in the xy coordinate system will also undergo transformation. Let the coordinates of P in the $x'y'$ system by (x', y') . We need a relationship between these coordinate systems. From the figure above,

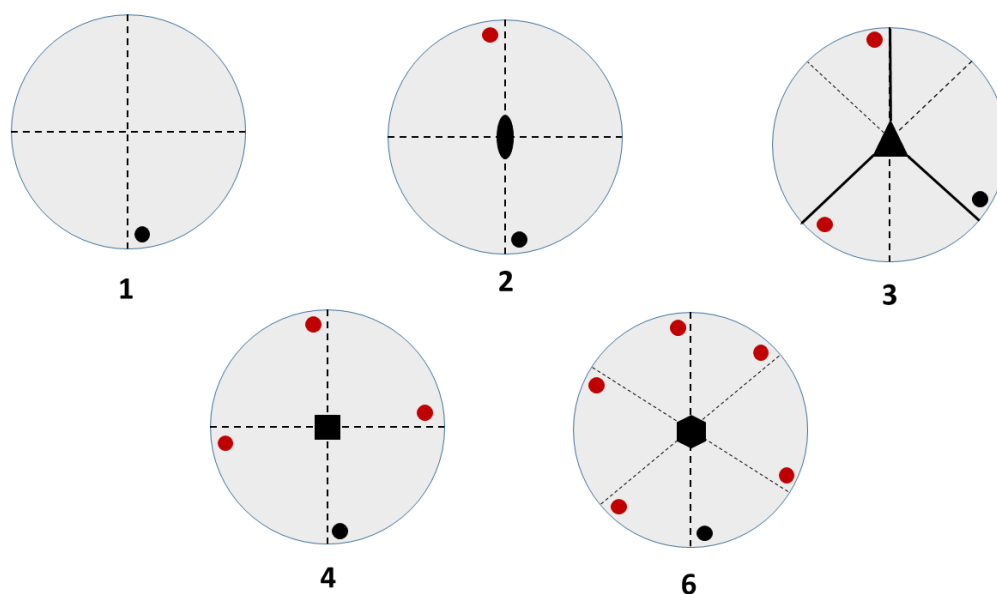
$$x' = OC + CD = x \cos \theta + y \sin \theta$$

$$y' = PB - DB = y \cos \theta - x \sin \theta$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = R(\theta) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

n-fold axis	Symbol	Orientation	Equivalent points	Transformation matrix
4	■	along a-axis	$(x, y, z), (x, \bar{z}, y),$ $(x, \bar{y}, \bar{z}), (x, z, \bar{y})$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$
		along b-axis	$(x, y, z), (\bar{z}, y, x)$ $(\bar{x}, y, \bar{z}), (z, y, \bar{x})$	$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$
		along c-axis	$(x, y, z), (\bar{y}, x, z),$ $(\bar{x}, \bar{y}, z), (y, \bar{x}, z)$	$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

It is instructive to calculate and determine the equivalent points for all other rotation axes. Generation of equivalent points due to operation of rotation axes can be depicted through stereograms (Fig. 4.zz). We will take on case and leave the rests to the reader for exercise.



Consider the case of 4-fold axis. Let us assume that the 4-fold axis is aligned to z-axis. Therefore, on the stereogram, equivalent points would appear at every 90° ($360^\circ/4$) angular separation. This has been shown in the Fig. 4.xx. Similar argument holds correct for 2, 3 and 6

fold axes where equivalent points are found at every 180°, 120°, and 60° angular separation, on the stereogram.

(v) *Inversion axes ($\bar{n}$)*: if an object, pattern or a lattice remains invariant after undergoing a rotation about an axis followed by an inversion through the center, it is said to possess inversion symmetry and the symmetry element representing it is called a rotation-inversion axis or simply inversion axis. Thus the operation of rotation-inversion is a combined operation involving symmetry with respect to an axis as well as a point. Inversion axis is represented by a bar/hat on the rotation axis ($\bar{n}$). Fig. 4.xx demonstrates the operation of a 2-fold inversion axis ($\bar{2}$).

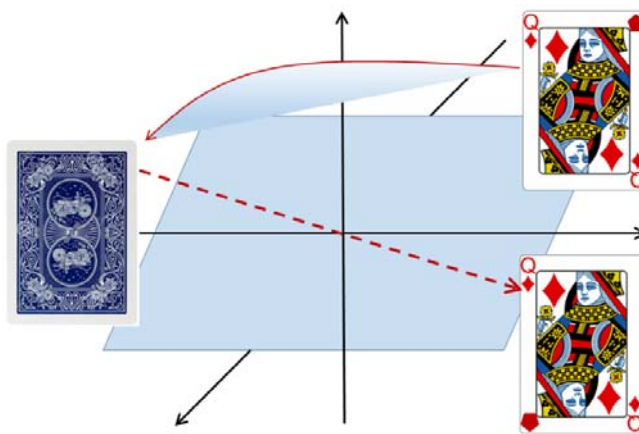


Fig. 4.xx:

Similar to the rotation axes, as shown in Table 4.3, inversion axes too have a set of graphical symbols to represent them. These are open symbols unlike their filled counterparts in Table 4.3. Table 4.xx lists the relevant symbols corresponding to the inversion axes.

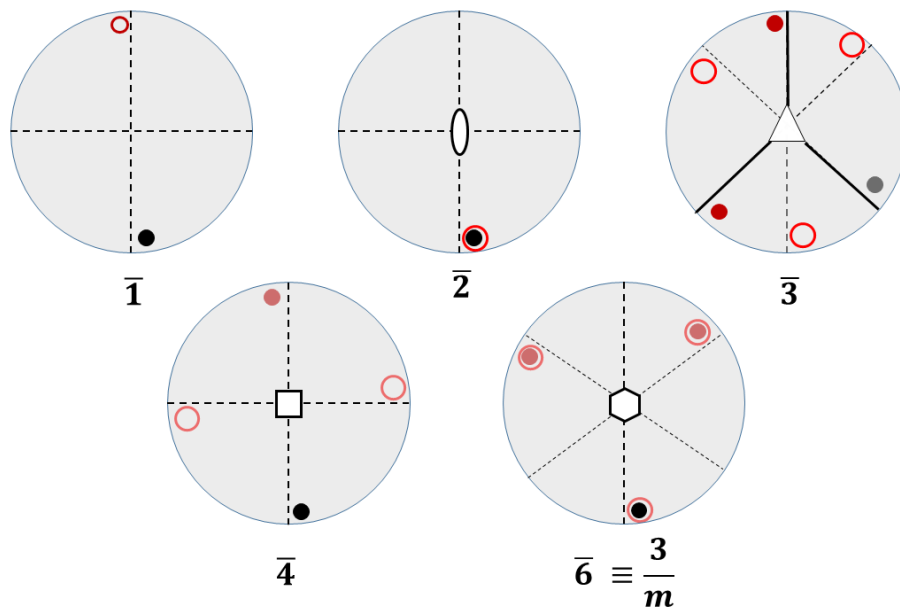
Table 4.xx

Sl No.	Inversion axis ($\bar{n}$ -fold)	Angle of rotation (θ)	Graphical symbol
1	$\bar{1}$	360°	○
2	$\bar{2}$	180°	∅
3	$\bar{3}$	120°	△
4	$\bar{4}$	90°	□
6	$\bar{6}$	60°	⬡

Similar to the earlier cases, the equivalent points generated due to the operation of the inversion axes will depend on the alignment of the rotation axis, involved. As discussed above, inversion axis is a combined operation of rotation followed by inversion. Therefore, the equivalent points generated could be determined considering the above two. Consider the generation of equivalent points due to $\bar{4}$. Let us assume the axis is aligned along the c-axis. A point (x,y,z) will undergo 4-fold rotation and then inversion. From our earlier discussion above, we know the matrix that transforms an initial point to new positions due to the operation of 4 and $\bar{1}$. Therefore, the product of these two transformation matrix should give us the transformation matrix to generate equivalent points for $\bar{4}$. Referring to Table 4.xx and 4.xx, we find that the transformation matrix for $\bar{4}$ that lies along c-axis, is:

$$T_{\bar{4}} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

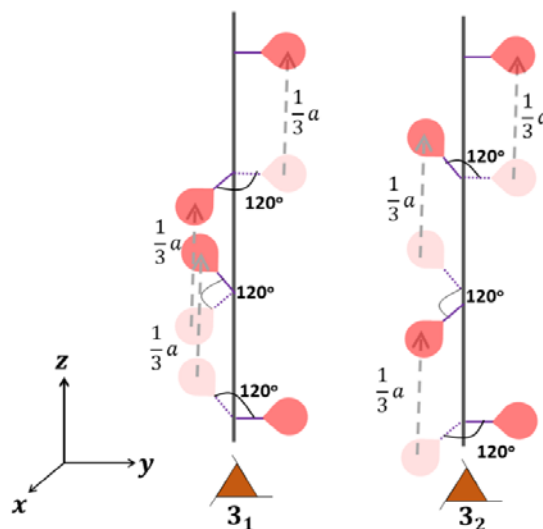
Thus, corresponding to a point (x,y,z), the equivalent points generated due to $\bar{4}$ are:



(vi) *Screw Axis*: Unlike the inversion axis where a combined rotation-inversion operation is performed, screw axes perform rotation followed by a translation. It is represented by n_x where

n stands for the rotation axis that can assume values, 2, 3, 4, and 6; x is used to mark the degree of translation (screw-pitch) along the axis of rotation. x can take values from 1 to $(n-1)$.

Let us see the following case, where we have screw axes based on 3-fold rotation axis. The value of x will be 1 and 2, as per above discussion and thus, we get two possible screw axes, viz., 3_1 and 3_2 . Fig. 4.yy demonstrates the generation of equivalent points due to the operation of the above two screw axes. In case of 3_1 , there is a 3-fold anti-clockwise rotation about the z -axis, followed by translation by $1/3$ lattice parameter. This operation generates the first equivalent point M' from the original point M . A second combined operation of 120° rotation-followed by $1/3^{\text{rd}}$ translation generates M'' . Another similar operation brings the original point translated by one lattice parameter. In case of 3_2 , followed by a 3-fold anti-clockwise rotation, there is a translation by $2/3$ lattice parameter which generates the point M' . Another operation of the above kind



Stereogram representation

(vii) Glide planes: Another coupled symmetry operation involving translational symmetry is glide-plane symmetry where reflection is followed by half-translation along a specified direction (a , b , c , n and d). Fig. 4.zz demonstrates the operation of a a -glide plane. The lattice with a a -glide plane will first undergo a reflection, perpendicular to a -axis and then a translation along the a -axis by $a/2$ distance where a is the lattice parameter along a -axis. Herein, the reader should note that for an orthogonal system, perpendicular to a -axis, there may lie either b or c axes.

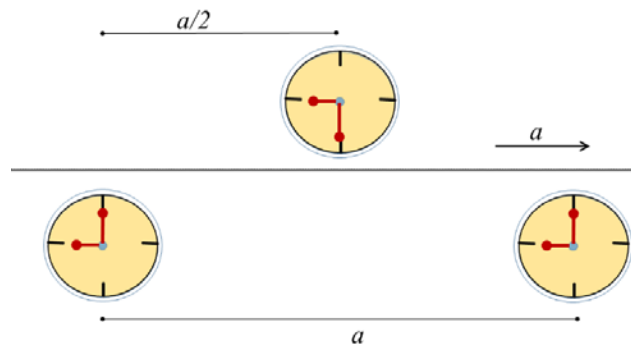


Fig. 4.zz: